Supplementary Materials for

"Specification and Estimation of Matrix Time Series Models with Multiple Terms"

Kurtuluş Kıdık[1,*] and Dietmar Bauer[1]

[1]Econometrics, Bielefeld University, Bielefeld, 33615, NRW, Germany
[*]Corresponding author, E-mail: kurtulus.kidik@uni-bielefeld.de

November 2025

This supplement is organized into three sections. Supplement S1 includes more details on estimation methods. Supplement S2 provides additional simulation results. Supplement S3 presents a table of critical values to different significance levels for the cointegration test.

S1 Estimators for the Multi-term MAR

S1.1 Deriving the ALSEs for the Multi-term MAR

If we assume that the error terms U_t are i.i.d. normal with a constant mean matrix and an arbitrary covariance matrix with $\Sigma_u = Cov(vec(U_t))$, estimators $\widehat{\boldsymbol{A}}^*$ and $\widehat{\boldsymbol{B}}^*$ are the solution of the least squares problem of $MAR_J(p)$ model (7) such that

$$\min_{\boldsymbol{A}^*,\boldsymbol{B}^*} \sum_t \|Y_t - \boldsymbol{A}^* X_t \boldsymbol{B}^{*\prime}\|_F^2 \tag{S1}$$

To minimize this least squares problem, we take partial derivatives of (S1) with respect to the entries of $\boldsymbol{A}^*$ and $\boldsymbol{B}^*$ and set them equal to zero. Arranging the corresponding equations in matrices yields the following first order conditions:

$$\sum_t \boldsymbol{A}^* X_t \boldsymbol{B}^{*\prime} \boldsymbol{B}^* X_t' - \sum_t Y_t \boldsymbol{B}^* X_t' = 0$$
$$\sum_t \boldsymbol{B}^* X_t' \boldsymbol{A}^{*\prime} \boldsymbol{A}^* X_t - \sum_t Y_t' \boldsymbol{A}^* X_t = 0$$

By solving these equations for the coefficient matrices $\boldsymbol{A}^*, \boldsymbol{B}^*$, we obtain the following system of equations:

$$\widehat{\boldsymbol{A}^*} = \left(\sum_t Y_t \widehat{\boldsymbol{B}}^* X_t'\right)\left(\sum_t X_t \widehat{\boldsymbol{B}}^{*\prime} \widehat{\boldsymbol{B}}^* X_t'\right)^{-1}$$
$$\widehat{\boldsymbol{B}^*} = \left(\sum_t Y_t' \widehat{\boldsymbol{A}}^* X_t\right)\left(\sum_t X_t' \widehat{\boldsymbol{A}}^{*\prime} \widehat{\boldsymbol{A}}^* X_t\right)^{-1}$$

As there is no analytical solution for this system of equations, it is iteratively solved by updating one part while keeping the others constant.

S1.2 Taking the Kronecker structure of the noise variance into account

Assume that the error terms U_t follow a matrix normal distribution, $U_t \sim \mathcal{N}_{M,N}(0, \Sigma_c \otimes \Sigma_r)$, such that the variance matrix equals a Kronecker product. Taking this structure into account, the conditional log-likelihood function of the multi-term MAR(p), $(MAR_J(p))$, model (7) is of the following form (up to constants)

$$l(\theta, \Sigma_r, \Sigma_c) \propto -\frac{1}{2}(T-p)(M \log|\Sigma_c| + N \log|\Sigma_r|) - \frac{1}{2}\sum_{t=p+1}^{T} \operatorname{tr}\left(\Sigma_r^{-1} R_t \Sigma_c^{-1} R_t'\right)$$

where, θ denotes the collection of parameters for $\boldsymbol{A}^*$ and $\boldsymbol{B}^*$ that are defined in Section-3.1. Further,

$$R_t = Y_t - \boldsymbol{A}^* X_t \boldsymbol{B}^{*\prime}$$

The first order conditions fulfilled by the maximum likelihood estimators (MLEs) of $\theta, \Sigma_r, \Sigma_c$ then take the following form:

$$\boldsymbol{A}^* \sum_t X_t \boldsymbol{B}^{*\prime} \Sigma_c^{-1} \boldsymbol{B}^* X_t' - \sum_t Y_t \Sigma_c^{-1} \boldsymbol{B}^* X_t' = 0$$
$$\boldsymbol{B}^* \sum_t X_t' \boldsymbol{A}^{*\prime} \Sigma_r^{-1} \boldsymbol{A}^* X_t - \sum_t Y_t' \Sigma_r^{-1} \boldsymbol{A}^* X_t = 0$$
$$M(T-p)\Sigma_c - \sum_t R_t' \Sigma_r^{-1} R_t = 0$$
$$N(T-p)\Sigma_r - \sum_t R_t \Sigma_c^{-1} R_t' = 0$$

By solving these equations for the coefficient and covariance matrices $\boldsymbol{A}^*, \boldsymbol{B}^*, \Sigma_r$ and Σ_c, we obtain the following system of four equations:

$$\widehat{\boldsymbol{A}^*} = \left(\sum_t Y_t \widehat{\Sigma}_c^{-1} \widehat{\boldsymbol{B}}^* X_t'\right) \left(\sum_t X_t \widehat{\boldsymbol{B}}^{*\prime} \widehat{\Sigma}_c^{-1} \widehat{\boldsymbol{B}}^* X_t'\right)^{-1}$$
$$\widehat{\boldsymbol{B}^*} = \left(\sum_t Y_t' \widehat{\Sigma}_r^{-1} \widehat{\boldsymbol{A}}^* X_t\right) \left(\sum_t X_t' \widehat{\boldsymbol{A}}^{*\prime} \widehat{\Sigma}_r^{-1} \widehat{\boldsymbol{A}}^* X_t\right)^{-1}$$

$$\widehat{\Sigma_c} = \frac{\sum_t \widehat{R}_t' \widehat{\Sigma}_r^{-1} \widehat{R}_t}{M(T-p)}$$
$$\widehat{\Sigma_r} = \frac{\sum_t \widehat{R}_t \widehat{\Sigma}_c^{-1} \widehat{R}_t'}{N(T-p)}$$

Given the initial values of the matrices $\boldsymbol{A}^*, \boldsymbol{B}^*, \Sigma_r$ and Σ_c, estimates are obtained by cyclically iteratively solving these four equations, as summarized in Algorithm S1.

Algorithm S1 Alternating optimization with separable covariance matrix (MLE).

Require: $\widehat{\boldsymbol{A}^*}_{(1)}, \hat{\Sigma}_{c,(1)}, \hat{\Sigma}_{r,(1)}$
  $i \leftarrow 1$
  while not converged **do**
    Calculate $\hat{X}_{t,(i)} \leftarrow \widehat{\boldsymbol{A}^*}_{(i)} X_t$
    Regress Y_t onto $\hat{X}_{t,(i)}$ to obtain $\widehat{\boldsymbol{B}^*}_{(i)}$ (taking $\hat{\Sigma}_{r,(i)}$ into account)
    Calculate $\tilde{X}_{t,(i)} \leftarrow X_t \widehat{\boldsymbol{B}^*}_{(i)}'$
    Regress Y_t onto $\tilde{X}_{t,(i)}$ to obtain $\widetilde{\boldsymbol{A}^*}_{(i)}$ (taking $\hat{\Sigma}_{c,(i)}$ into account)
    Calculate $\hat{\hat{\Phi}}_{j,(i)}, j = 1, ..., p$ corresponding to $\widetilde{\boldsymbol{A}^*}_{(i)}, \widehat{\boldsymbol{B}^*}_{(i)}$.
    $i \leftarrow i + 1$
    Refactor $\mathcal{R}(\hat{\hat{\Phi}}_{j,(i-1)})$ to obtain $\widehat{\boldsymbol{A}^*}_{(i)}, \widehat{\boldsymbol{B}^*}_{(i)}$.
    Calculate residuals $\hat{R}_t = Y_t - \widehat{\boldsymbol{A}^*}_{(i)} X_t \widehat{\boldsymbol{B}^*}_{(i)}'$.
    $\hat{\Sigma}_{c,(i)} \leftarrow \frac{1}{M(T-p)} \sum_{t=1}^{T} \hat{R}_t' \hat{\Sigma}_{r,(i-1)}^{-1} \hat{R}_t$
    $\hat{\Sigma}_{r,(i)} \leftarrow \frac{1}{N(T-p)} \sum_{t=1}^{T} \hat{R}_t \hat{\Sigma}_{c,(i)}^{-1} \hat{R}_t'$
  end while

S1.3 Quasi Maximum Likelihood estimation

We discussed the theoretical background of the quasi-MLEs in Section 3.2. There are many nonlinear optimization algorithms for estimating quasi maximum likelihood. For example, any steepest descent methods or their derivatives, such as gradient descent/ ascent methods, could be used. In our research, we use the MATLAB functions $fminunc$ for optimization. The link for related codes can be found at the Appendix B in the main text. https://github.com/kurtuluskidik/SEMaTS.git.

S2 Additional Simulation Results

This section provides additional results for the simulations discussed in Section 6 of the main text. The purpose of these supplementary results is to offer a more comprehensive analysis of the performance and robustness of our proposed approaches. By providing this additional information, we aim to strengthen the overall validity and reliability of our research findings. We also consider the following two cases that we discussed in the data generating process in our main text. To remember different covariance settings from the main text, let the noise term be i.i.d. normal with covariance matrix $\Sigma_u = Cov(vec(U_t))$. We consider two covariance structures for the noise process U_t:

- Case-I: The generic error covariance matrix without any structure: $\Sigma_u = P\Lambda P'$, where the elements of the diagonal matrix Λ are drawn by i.i.d. absolute standard normal random variables and P is an orthonormal random matrix of eigenvectors. This setting ensures that Σ_u is a symmetric positive define matrix.

- Case-II: Separable covariance structure with Kronecker product form: $\Sigma_u = \Sigma_c \otimes \Sigma_r$, where Σ_c, column-wise covariance matrix, and Σ_r, row-wise covariance matrix, are generated similarly to Σ_u in case-I.

Figure S1 presents a comparative performance analysis of the corresponding estimators under Case-II. In general, all multi-term estimators perform reasonably well, exhibiting reduced variance and bias with increasing sample size (T), as seen from left to right in each row and sub-figures display a similar pattern to the Figure-1 in the main text. However, we observe the effect of the covariance structure that supports the matrix form. Thus, while the VAR estimator consistently shows the worst performance, the single-term estimators of MAR demonstrate improvements over VAR, though they exhibit more bias and variability than their multi-term counterparts with increasing complexity. Now, in the case of increasing complexity with a small sample size MLE estimates outperform $qMLE$.

Across all variations, VAR consistently shows higher error and variability, especially in low T with complex scenarios. As T increases, performance improves across all estimators, but the multi-term models generally outperform their single-term counterparts. Among them, $qMLE$ and MLE tend to yield the lowest error and variability, especially in high-dimensional settings and for large T, suggesting their robustness and superior accuracy for complex matrix time series data.

The general results of two figures are that all multi-term MAR estimators clearly outperform the stacked VAR estimator and single-term MAR estimators. Second, it can be observed that as the sample size and dimensions increase, the error of the estimation coefficient decreases accordingly, and multi-term $qMLE$ is consistently the most accurate and stable estimator, which confirms the consistency of our proposed estimators.

The comparison between Figure 2 in the main text and Figure S2 underscores the substantial impact of sample size on the predictive accuracy and stability of matrix autoregressive estimators under Case-I. When the sample size increases from ($T = 100$ to $T = 500$), all estimators exhibit markedly reduced variability in $RatioMSPE$, with tighter boxplots and fewer outliers. This improvement is most pronounced for the quasi-maximum likelihood estimator with multi-term structure, $qMLE_J(p)$, which consistently achieves superior performance relative to the VAR benchmark, particularly for longer prediction horizons and higher-dimensional settings.

While the relative ranking of estimators remains consistent across both sample sizes, with multi-term estimators ($ALSE_J(p), MLE_J(p), qMLE_J(p)$) outperforming single-term ones, and $ALSE(p)$ yielding the poorest performance, the performance gaps become more distinct as sample size increases. Moreover, both figures confirm that relative predictive performance improves with increasing prediction horizon (T_v) and matrix dimension (M, N), though these effects are more reliably observed with larger sample sizes.

For Section 6.2, additional simulation results of using the sample adjusted critical values is presented in Table S1. Critical values from the fourth to the eight column of the Table-A.3 in the main text have been used.

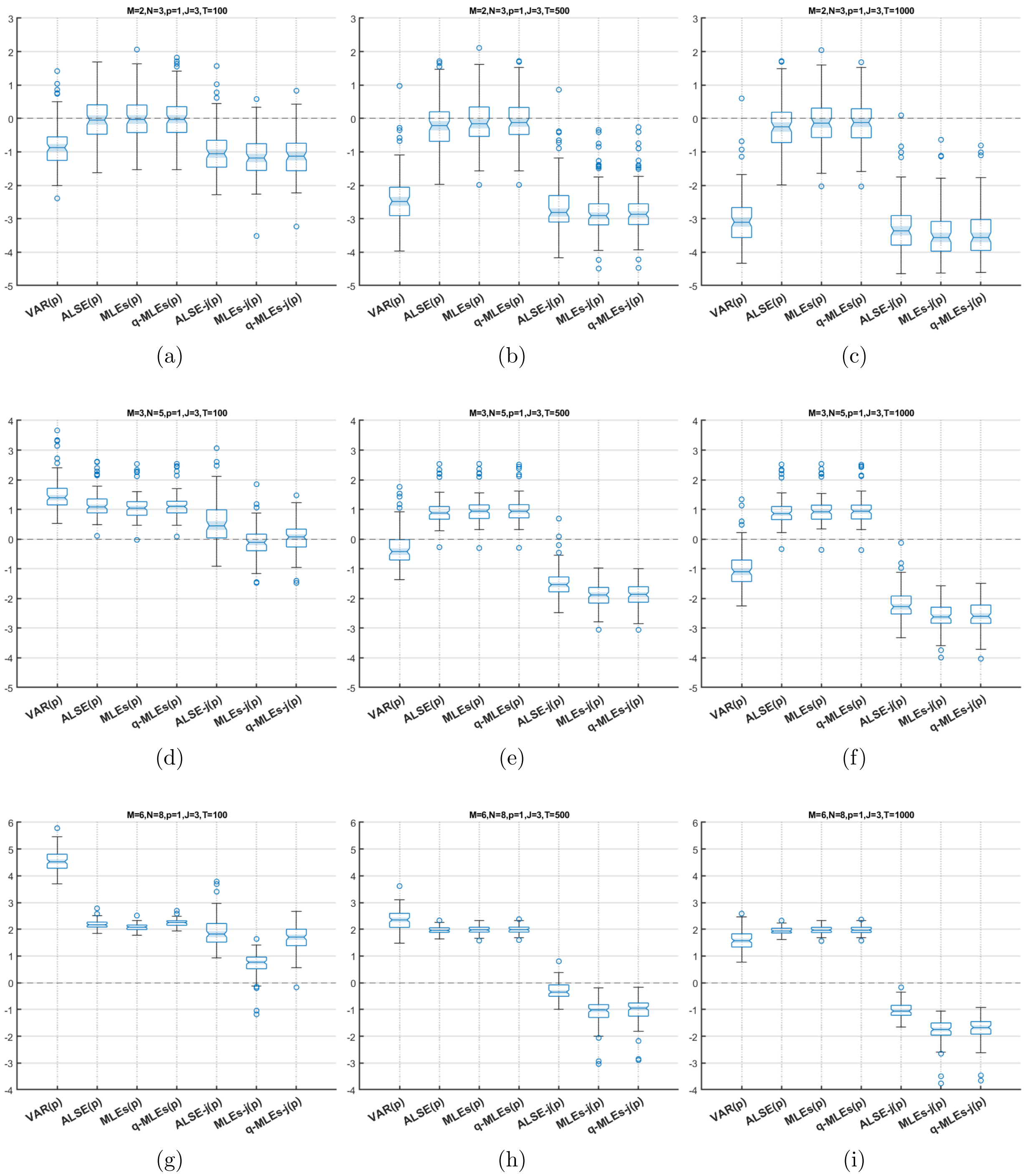

Figure S1: Comparison of estimators under Case-II. $VAR(p)$, $ALSE(p)$, $MLEs(p)$, $qMLE(p)$, $ALSE_J(p)$, $MLE_J(p)$, $qMLE_J(p)$ (from left to right in each plot). Rows correspond to $(M, N) = (2, 3), (3, 5)(6, 8)$ and columns correspond to $T = 100, 500, 1000$.

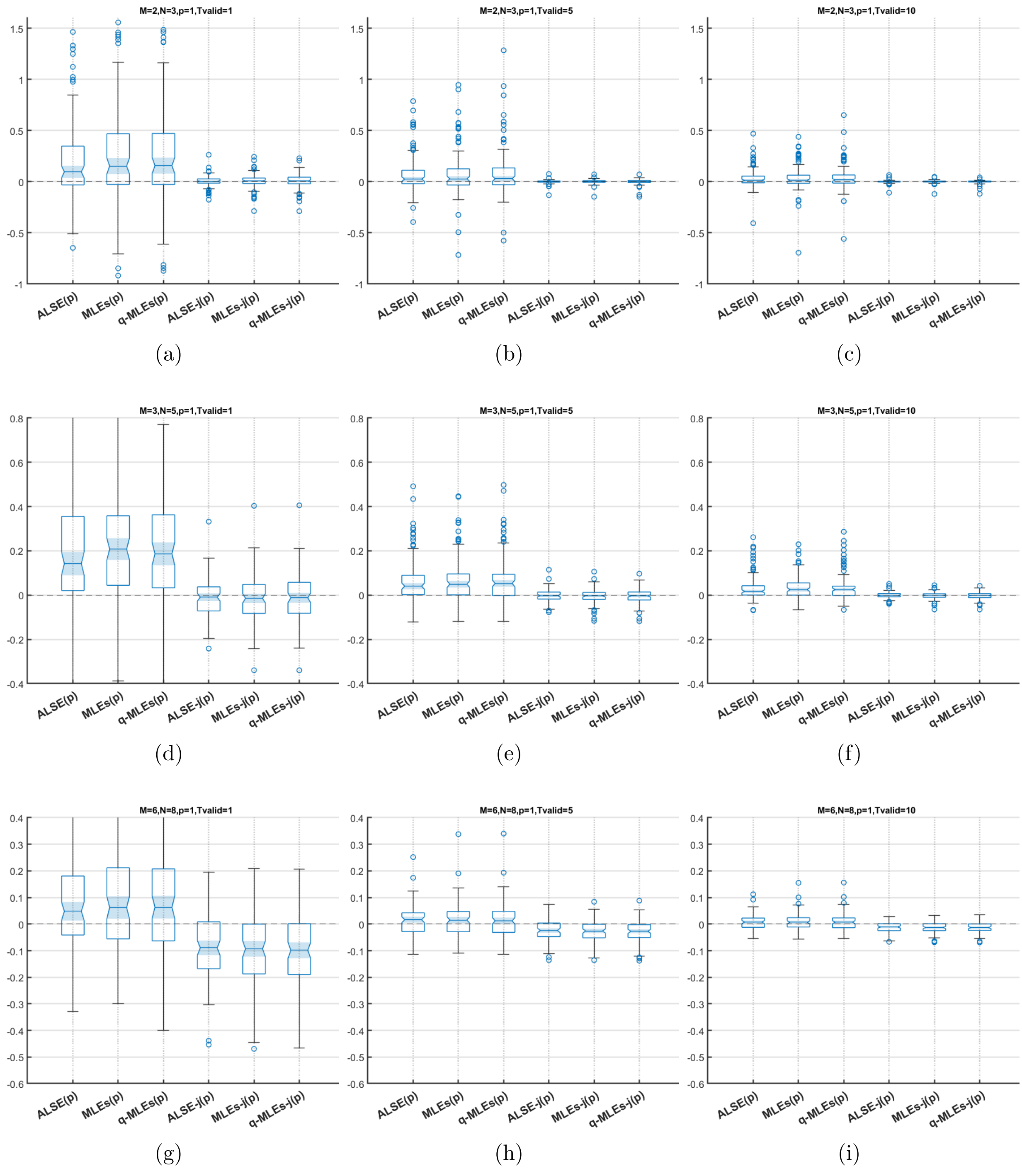

Figure S2: RatioMSPE under Case-I second column of Figure 2 in the main text, (1b, 1e, 1h), $T = 500$. $ALSE(p)$, $MLE(p)$, $qMLE(p)$, $ALSE_J(p)$, $MLE_J(p)$, $qMLE_J(p)$ (from left to right in each plot) with $VAR(p)$ is a benchmark. Rows correspond to $(M, N) = (2, 3), (3, 5)(6, 8)$ and columns correspond to prediction horizon $T_v = 1, 5, 10$ increases from left to right.

True Rank	Method	T	Min. Rank	Median Rank	Max. Rank	Ave. Rank	S.D.	Success (percent)
0	EC-MAR	(200)	0	0	3	0.19	0.52	86 %
	VECM	(200)	0	0	6	0.14	0.72	95 %
	EC-MAR	(500)	0	0	2	0.15	0.45	89 %
	VECM	(500)	0	0	1	0.05	0.21	95 %
1	EC-MAR	(200)	0	1	4	1.08	0.61	70 %
	VECM	(200)	0	0	4	0.22	0.69	8 %
	EC-MAR	(500)	1	1	2	1.20	0.40	80 %
	VECM	(500)	0	1	3	0.86	0.51	75 %
6	EC-MAR	(200)	3	5	7	5.19	0.69	28 %
	VECM	(200)	0	3	9	2.98	2.23	8 %
	EC-MAR	(500)	4	6	6	5.68	0.48	69 %
	VECM	(500)	4	5	7	5.50	0.64	43 %
11	EC-MAR	(200)	7	9	11	9.25	0.93	5 %
	VECM	(200)	2	8	15	7.91	2.39	13 %
	EC-MAR	(500)	9	11	12	10.60	0.56	55 %
	VECM	(500)	9	10	15	10.48	0.87	40 %
16	EC-MAR	(200)	11	15	20	14.72	1.31	29 %
	VECM	(200)	6	14	19	13.81	2.21	12 %
	EC-MAR	(500)	14	16	17	15.96	0.37	89 %
	VECM	(500)	14	16	17	15.81	0.52	72 %
	EC-MAR	(200)	14	19	20	18,50	0.84	54 %
19	VECM	(200)	14	19	20	18.35	1.04	48 %
	EC-MAR	(500)	18	19	20	19.01	0.22	95 %
	VECM	(500)	18	19	20	19.03	0.26	93 %

Table S1: Comparison of vECMAR and VECM estimations with cointegration rank selection for dimensions $M = 4$ and $N = 5$ data under covariance structure is Case-I. Simulation results are from $T = 200$ and $T = 500$ with no cointegration ($r = 0$) and 5 rank cases ($r = 1, 6, 11, 16, 19$). The last column shows success rate over 100 repetitions. Critical values from the fourth to the eight columns of the Table-A.3 in the main text have been used.

S3 Critical values

This section provides critical values for additional asymptotic percentiles—namely 90%, 95%, 97.5% and 99%.

Table S2: Critical values for trace test for cointegration rank with no constant and deterministics

p-r	90.0%	95.0%	97.5%	99.0%
1	3.0100	4.1310	5.5091	7.1582
2	10.4773	12.2658	14.0356	16.1747
3	21.8622	24.5949	26.8339	29.8188
4	37.3445	40.3621	43.2011	47.0336
5	56.4766	60.2090	63.4930	67.3903
6	79.3210	83.6911	87.9624	93.2362
7	106.9658	111.8609	116.2983	121.9886
8	138.4017	144.4176	150.1425	155.6891
9	173.6605	179.4645	185.1517	191.8989
10	213.2526	220.1860	227.0293	233.4213
11	255.8442	263.6575	270.6760	279.3493
12	303.8350	311.6003	318.9930	326.3640
13	354.7356	362.9293	370.6955	380.0321
14	409.6714	419.1031	427.4947	436.9744
15	469.1357	479.4069	487.6135	498.0038
16	532.5796	542.6171	553.2051	563.3654
17	600.8269	612.1593	621.5791	632.8025
18	671.3105	683.4235	693.5222	704.2964
19	748.0202	761.2085	771.6292	782.6004
20	825.8168	839.0941	850.4799	863.5438